TASK 3: WEBSITE TRAFFIC ANALYSIS

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GITHUB LINK: https://github.com/Divyabharathi42/Website-Traffic-Analysis

Objective

The primary objective of this project is to analyze website traffic data and understand user engagement patterns. The analysis focuses on identifying the most accessed links, evaluating traffic distribution across countries, analyzing artist and track popularity, and deriving actionable insights that can improve website performance and user experience.

Dataset

The dataset consists of website traffic records containing user interaction events. Key attributes include:

- **Date** – Timestamp of user activity.
- **Event** – Type of interaction performed by the user.
- **Link ID** – Unique identifier for accessed links.
- **Track** – Associated track information.
- **Artist** – Artist linked to the content.
- **Country** – User's geographic location.

Data Preprocessing

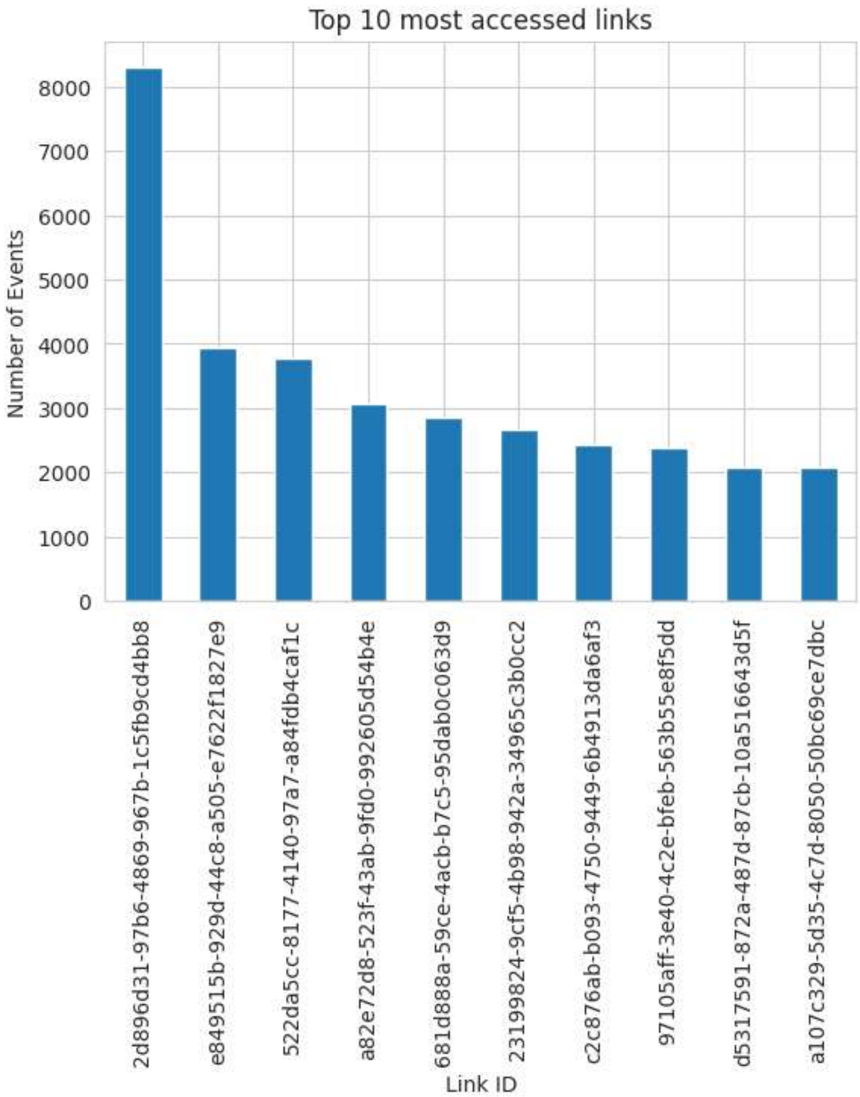
- Converted date fields into proper datetime format.
- Removed duplicate records.
- Removed rows with missing values in critical columns such as Event, Date, and Link ID.
- Performed data quality checks and validation.

Tools Used

- **Google Colab** – Development and execution environment.
- **Python** – Programming language used for analysis.
- **Pandas** – Data manipulation and preprocessing.
- **NumPy** – Numerical computations.
- **Matplotlib** – Data visualization.
- **Seaborn** – Statistical visualization and chart styling.
- **Plotly Express** – Interactive visualizations.

Visualizations Included

- Top 10 most accessed links



Business Recommendations

1. Promote High-Traffic Links

Identify the most accessed links and prioritize them on landing pages to improve user engagement and conversions.

2. Optimize Content Strategy

Analyze the most popular artists and tracks to understand audience preferences and create similar content that attracts more visitors.

3. Focus on High-Performing Regions

Use country-level traffic insights to target marketing campaigns and localized content toward regions generating the highest engagement.

4. Improve User Journey

Study event flow patterns to identify drop-off points and optimize navigation paths, reducing friction in the user experience.

5. Monitor Traffic Trends Regularly

Establish periodic traffic monitoring dashboards to track changes in user behavior and quickly respond to emerging trends.

Conclusion

The Website Traffic Analysis project successfully explored user interaction data to uncover valuable insights about website usage patterns. Through data cleaning, exploratory analysis, and visualization, key metrics such as popular links, leading countries, and top-performing artists/tracks were identified. The findings can help businesses improve content strategy, enhance user engagement, optimize marketing efforts, and support data-driven decision-making for future growth.